

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING****Product Identifier**

**Perihal Produk:** BHI Broth w/Hemin, Vit K  
**Product Description:** BHI Broth w/Hemin, Vit K  
**Cat No. :** R060276, R060278, R07019

**Relevant identified uses of the substance or mixture and uses advised against**

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

**Details of the supplier of the safety data sheet**

**Company** Thermo Scientific Microbiology Sdn Bhd  
No.6, Jalan TTC 6, Taman Teknologi Cheng,  
Cheng, 75250 Melaka, Malaysia  
+606 334 0975 .

**Supplier** Remel  
12076 Santa Fe Drive Lenexa,  
KS 66215 United States  
Telephone: 1-800-255-6730  
Fax:1-800-621-8251

**E-mail address** mbd-sds@thermofisher.com

**Emergency Telephone Number**

(603) 5122 8888  
CHEMTREC Malaysia **1-800-815-308** (Malay)  
CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION****Classification of the substance or mixture****Label Elements****Hazard Statements****Precautionary Statements****Other Hazards**

This product does not contain any known or suspected endocrine disruptors

**SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS**

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| Component                                                                                                                                        | CAS No     | Weight % |
|--------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------|
| Ferrate(2-),<br>chloro[7,12-diethenyl-3,8,13,17-tetramethyl-21H,23H-porphine-2,18-dipropano<br>ato(4-)-N21,N22,N23,N24]-, dihydrogen, (SP-5-13)- | 16009-13-5 | 0.09     |
| Non-Hazardous Components                                                                                                                         | NA         | 3.56     |
| Water                                                                                                                                            | 7732-18-5  | 96.27    |
| Vitamin K1 (Phytylmenadione)                                                                                                                     | 84-80-0    | 0.09     |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

|                                           |                                                                                                                                                  |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                        | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a physician.                             |
| <b>Skin Contact</b>                       | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes.                                            |
| <b>Ingestion</b>                          | Clean mouth with water and drink afterwards plenty of water.                                                                                     |
| <b>Inhalation</b>                         | Remove to fresh air.                                                                                                                             |
| <b>Self-Protection of the First Aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |

### Most important symptoms and effects, both acute and delayed

No information available.

### Indication of any immediate medical attention and special treatment needed

**Notes to Physician** Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### **Suitable Extinguishing Media**

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors.

#### **Hazardous Combustion Products**

None under normal use conditions.

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

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## Personal Precautions, Protective Equipment and Emergency Procedures

Ensure adequate ventilation.

## Environmental precautions

See Section 12 for additional Ecological Information.

## Methods and Material for Containment and Cleaning Up

Soak up with inert absorbent material.

## Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Ensure adequate ventilation.

### Conditions for Safe Storage, Including any Incompatibilities

Keep container tightly closed in a dry and well-ventilated place.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component                                                                                                                                               | Malaysia | ACGIH TLV                | OSHA PEL                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------|------------------------------------|
| Ferrate(2-),<br>chloro[7,12-diethenyl-3,8,13,17-tetra<br>methyl-21H,23H-porphine-2,18-dipr<br>opanoato(4-)-N21,N22,N23,N24]-,<br>dihydrogen, (SP-5-13)- |          | TWA: 1 mg/m <sup>3</sup> | (Vacated) TWA: 1 mg/m <sup>3</sup> |

| Component                                                                                                                                               | European Union | The United Kingdom                                                | Germany |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-------------------------------------------------------------------|---------|
| Ferrate(2-),<br>chloro[7,12-diethenyl-3,8,13,17-tetra<br>methyl-21H,23H-porphine-2,18-dipr<br>opanoato(4-)-N21,N22,N23,N24]-,<br>dihydrogen, (SP-5-13)- |                | STEL: 2 mg/m <sup>3</sup> 15 min<br>TWA: 1 mg/m <sup>3</sup> 8 hr |         |

### Exposure Controls

#### Engineering Measures

Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

#### Personal protective equipment

##### Eye Protection

Wear safety glasses with side shields (or goggles)

##### Hand Protection

Protective gloves

##### Skin and body protection

Long sleeved clothing

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Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.  
(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

## Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

When RPE is used a face piece Fit Test should be conducted

## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

## Environmental exposure controls

No information available

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

#### Appearance

#### Physical State

Liquid

#### Odor

No information available

#### Odor Threshold

No data available

#### pH

Not applicable

#### Melting Point/Range

No data available

#### Softening Point

No data available

#### Boiling Point/Range

No information available

#### Flash Point

No information available

Method - No information available

#### Evaporation Rate

No data available

#### Flammability (solid,gas)

No information available

#### Explosion Limits

No data available

#### Vapor Pressure

No data available

#### Vapor Density

No data available

(Air = 1.0)

#### Specific Gravity / Density

No data available

#### Bulk Density

No data available

#### Water Solubility

No information available

#### Solubility in other solvents

No information available

#### Partition Coefficient (n-octanol/water)

#### Autoignition Temperature

No data available

#### Decomposition Temperature

No data available

#### Viscosity

No data available

#### Explosive Properties

No information available

#### Oxidizing Properties

No information available

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## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

### Chemical Stability

Stable under normal conditions.

### Possibility of Hazardous Reactions

**Hazardous Polymerization** No information available.  
**Hazardous Reactions** No information available.

### Conditions to Avoid

None known.

### Incompatible Materials

None known.

### Hazardous Decomposition Products

None under normal use conditions.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### Acute Toxicity

| Component                    | LD50 Oral                  | LD50 Dermal | LC50 Inhalation |
|------------------------------|----------------------------|-------------|-----------------|
| Water                        | -                          | -           | -               |
| Vitamin K1 (Phytolmenadione) | LD50 > 33487 mg/kg ( Rat ) |             |                 |

#### Chronic Toxicity Carcinogenicity

There are no known carcinogenic chemicals in this product

**Sensitization**  
**Mutagenic Effects**  
**Reproductive Effects**  
**Developmental Effects**  
**Target Organs**

No information available  
No information available  
No information available  
No information available  
No information available.

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## SECTION 12: ECOLOGICAL INFORMATION

|                                      |                                                                                                                             |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <u>Ecotoxicity effects</u>           | Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants. |
| <u>Persistence and degradability</u> | No information available                                                                                                    |
| <u>Bioaccumulative potential</u>     | No information available                                                                                                    |
| <u>Mobility in soil</u>              | No information available.                                                                                                   |
| <u>Other adverse effects</u>         | No information available                                                                                                    |

## SECTION 13: DISPOSAL CONSIDERATIONS

|                                            |                                                                                               |
|--------------------------------------------|-----------------------------------------------------------------------------------------------|
| <u>Waste treatment methods</u>             |                                                                                               |
| <u>Waste from Residues/Unused Products</u> | Dispose of in accordance with local regulations                                               |
| <u>Contaminated Packaging</u>              | Empty containers should be taken to an approved waste handling site for recycling or disposal |

## SECTION 14: TRANSPORT INFORMATION

|                                     |                                 |
|-------------------------------------|---------------------------------|
| <u>IMDG/IMO</u>                     | Not regulated                   |
| <u>Road and Rail Transport</u>      | Not regulated                   |
| <u>IATA</u>                         | Not regulated                   |
| <u>Special Precautions for User</u> | No special precautions required |

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

|                                  |            |
|----------------------------------|------------|
| <u>International Inventories</u> | X = listed |
|----------------------------------|------------|

| Component                                                                                                                 | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL |
|---------------------------------------------------------------------------------------------------------------------------|-----------|------|-----|-------|------|------|-------|------|------|
| Ferrate(2-),<br>chloro[7,12-diethenyl-3,8,13,17-tet<br>ramethyl-21H,23H-porphine-2,18-<br>dipropanoato(4-)-N21,N22,N23,N2 | 240-140-1 | X    | X   | -     | X    | X    | X     | -    | -    |

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|                               |           |   |   |   |   |   |   |   |          |
|-------------------------------|-----------|---|---|---|---|---|---|---|----------|
| 4]-, dihydrogen, (SP-5-13)-   |           |   |   |   |   |   |   |   |          |
| Water                         | 231-791-2 | X | X | X | X |   | X | X | KE-35400 |
| Vitamin K1 (Phytilymenadione) | -         | X | X | X | X | X | X | X | KE-24853 |

## National Regulations

**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

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Revision Summary

Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

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**End of Safety Data Sheet**